

**SURPRISE! USING
PHYSIOLOGICAL STRESS FOR
ALLOSTATIC REGULATION
UNDER THE ACTIVE
INFERENCE FRAMEWORK**



CORTISOL SERVES TWO KEY ALLOSTATIC FUNCTIONS:

- **Adjusting the Homeostatic Set Point:** Cortisol calibrates the prior target value of a life-essential physiological variable, helping the agent adjust its physiology to meet perceived environmental demands.
- **Modifying Learning:** Cortisol influences the rate at which the agent learns about the statistical contingencies of its environment (i.e., learning of its generative model).g Rates



AGENT SIMULATION

- **A. Simulation Environment** (proposed models in a computational environment)
- **Components:**
 - **Agent**(an artificial physiology (a “body”)) It has an artificial body regulated by homeostatic (maintaining stability) and allostatic (anticipating changes) control. It also has a brain-like generative model that helps it make inferences and choose actions to stay alive.
 - probabilistic (generative) model (its “brain”)
 - **External Environment**
 - two types of resources—food and friend

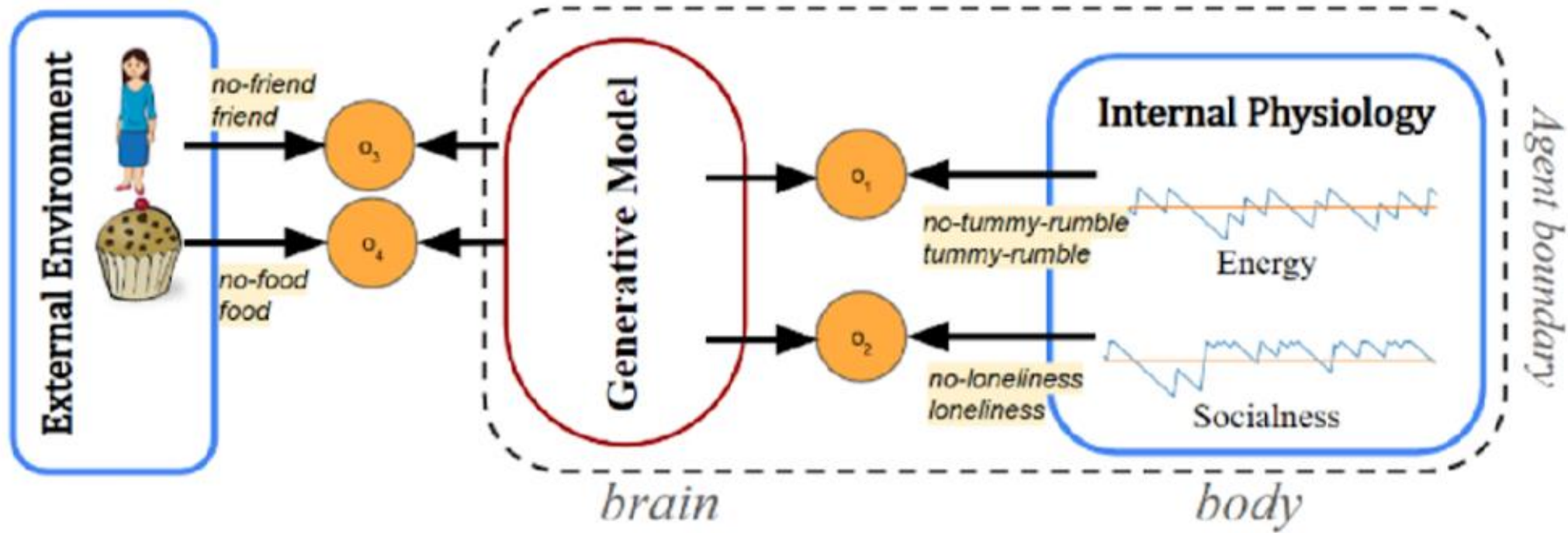
The agent must infer the (hidden) state (its motivation state) that it believes causes the interoceptive (and exteroceptive) signals that it observes



- The agent's goal is to stay alive by keeping its internal variables (like energy and socialness) above certain levels. It does this by interpreting internal signals (like hunger or loneliness) and taking actions to correct any imbalances.



DISEMBODIED, COMPUTATIONAL AGENT



▪ **External Environment**

- world from which the agent can receive observations and interact with.
- two resources (food and friend): 1 if they are observed, else 0

▪ **Agent Physiology and Action**

- Internal Variables: Energy (v1) and Socialness (v2)(ranging from 0 to 1)
- homeostatic set point d (initially set to 0.7), The agent needs to keep these values above the set points (0.7) to stay alive. Like biological organisms, these variables incur small deficits over time, moving away from ideal (attractor) states and towards states that threaten viability and life (i.e. running out of Energy):
- The agent loses a small amount (0.03) of both Energy and Socialness at each time step.
- The agent needs to keep these values above the set points (0.7) $v_n \geq d_v$ to stay alive. through either error-correcting action (homeostasis), or anticipatory/predictive adjustments (allostasis). If Energy drops to 0, the agent dies.

$$v_{n,t} = v_{n,t-1} - \gamma_n$$

value of internal variable v

rate of loss



SIGNALS AND ACTIONS

- If the internal variables fall below their set points, the agent receives internal signals: "**tummy rumble**" for low Energy and "**loneliness**" for low Socialness.
- To address these needs, the agent can choose from three actions: **eat**, **play**, or **explore**.
- The agent selects actions based on minimizing future prediction errors (i.e., choosing actions that it believes will best meet its needs in the future).



GENERATIVE MODEL

- This model helps the agent understand the world and decide what actions to take to maintain its internal states (like Energy and Socialness).
- The model is a Partially-Observable Markov Decision Process (POMDP), which includes:
 - **Observations (o):** What the agent senses.
 - **Hidden States (s):** The underlying causes of these observations (not directly observable).
 - **Actions (u):** The actions the agent can take.



Model Parameter	Formal definition	Description
A Matrix	$p(o s)$	Sensory likelihood mapping: the probability distributions that each hidden state s caused observation o , i.e. <i>“what is the likelihood that each motivation state s caused the interoceptive and exteroceptive data o that I have observed?”</i>
B Matrix	$p(s_t s_{t-1}, u_{t-1})$	Transition distribution describing the probability of transitioning to a state s at time t given state s and action u at $t-1$, i.e. <i>“what is the likelihood that my motivation state is s given (my beliefs) about the state I was just in and the action that I just took?”</i>
C Vector	$p(o C)$	Set of preferences for each observation, i.e. <i>“what sensory (interoceptive/exteroceptive) signals do I prefer to observe?”</i>



- **Observations**

- two **interoceptive** (Interoceptive signals are generated from the agent's physiology)
 - **Tummy and lonely**
- **Exteroceptive** (exteroceptive signals are generated by the external world)
 - presence of **food or a friend**

- **Hidden State**

- single hidden state: motivational state
 - **Hungry** : has a high probabilistic cause of observing food and/or tummy-rumble
 - **Playful** : likely cause of observing loneliness and/or friend
 - **Satisfied** : likely cause of observing notummy-rumble and no-loneliness.

$p(o|s)$

Tummy	Hungry
food	playful
lonely	satisfied
friend	



- **Action selection**

- **Process:** The agent selects actions to minimize future prediction errors (surprisal) and match its internal preferences.
- **Goal:** To maintain homeostasis by predicting which actions will most likely lead to preferred sensory signals (no "**tummy-rumble**" and no "**loneliness**").

Eat: Consumes food to increase Energy.

Play: Engages with a friend to increase Socialness.

Explore: Searches the environment for resources (guarantees observation of both food and friend in the next step but does not directly change internal states).

Each of the two consummatory behaviours, eat and play, satisfy the internal variables Energy and Social, by a value of +0.4: explore does not contribute to changes in internal variables.



- **Learning**

- Updating the Generative model
- Learning Rate: Initially set to 0.05 and influenced by cortisol levels.

Observation Modalities	$o^{Tummy} \in \{ no-tummy-rumble, tummy-rumble \}$ $o^{Lonely} \in \{ no-loneliness, loneliness \}$ $o^{food} \in \{ no-food, food \}$ $o^{friend} \in \{ no-friend, friend \}$
Hidden States	$s^{motivation} \in \{ hungry, playful, satisfied \}$
Actions	$u^{behaviour} \in \{ eat, play, explore \}$



Simulating Cortisol Secretion and Allostatic Functions

Cortisol secretion in this model is influenced by two main factors:

Changes in Prediction Error (Surprisal): that is, it is a measure of the difference between predicted observations and the actual sensory inputs at a given time step.

Uncertainty over Action Selection: We define uncertainty here as the difference between the best and worst actions (in terms of minimizing expected free energy), i.e. indecisiveness of how to best minimize future prediction errors.

$$CT_t = CT_{t-1} + \boxed{(-\ln(p(o_t)) - -\ln(p(o_{t-1})))} + \boxed{(1 - (G_{max} - G_{min}))} (2)$$

Represents the change in surprisal between the current and previous time steps

Represents the normalized difference between the maximum and minimum posterior probabilities over actions, indicating the level of uncertainty.



Allostatic Functions of Cortisol

Adjusting Internal Homeostatic Set Point:(i.e. the internal value associated with the prior preference $p(o | C)$) associated with the life-critical variable Energy:

$$d_{energy, t} = d_{energy, t-1} / (1 + CT_t - CT_{t-1})$$

set point of Energy at time t

Modulating the Learning Rate for State Transition Probabilities:On a slower time scale, cortisol modulates the learning rate used for updating state transition probabilities in the generative model.

This modulation is akin to Hebbian learning and represents cortisol's effect on synaptic plasticity and learning.

The formula for updating the B matrix is:

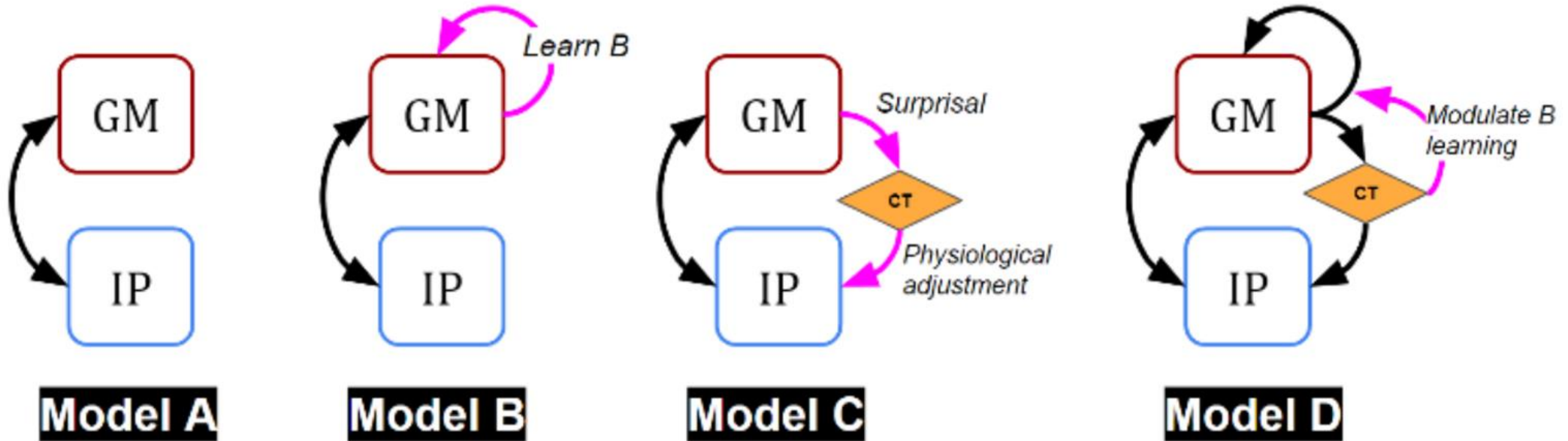
$$P(B) = B + (\lambda \times (1 - CT))$$

posterior B matrix

predefined learning parameter (0.05)



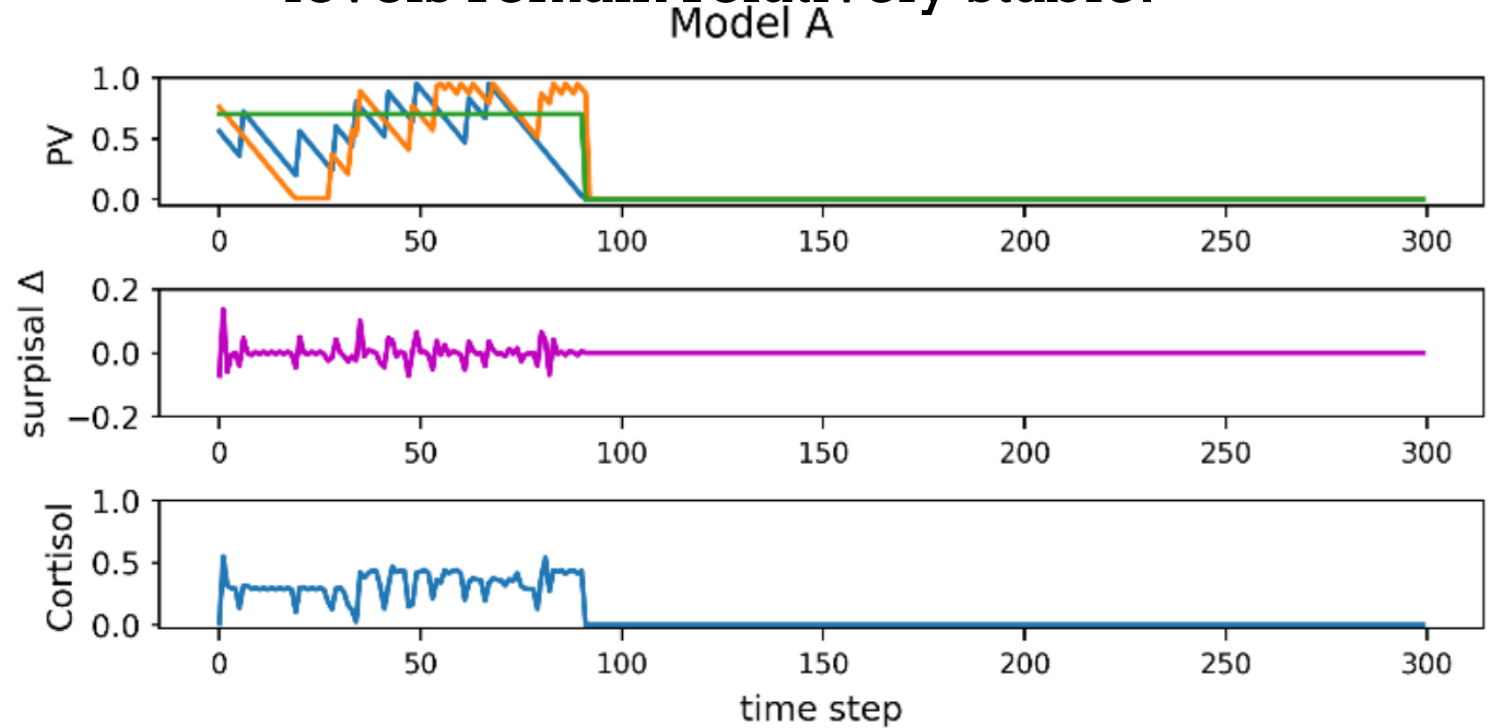
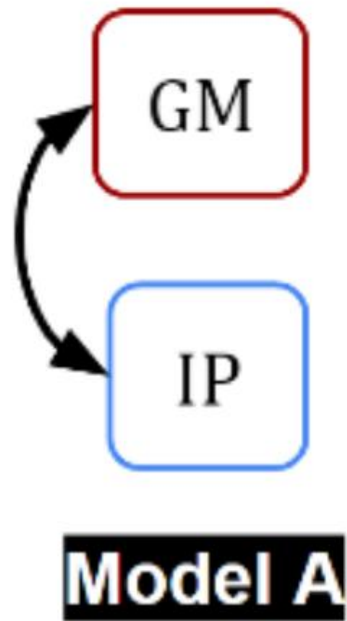
different models to explore various allostatic effects of cortisol



Simple Homeostatic Regulation

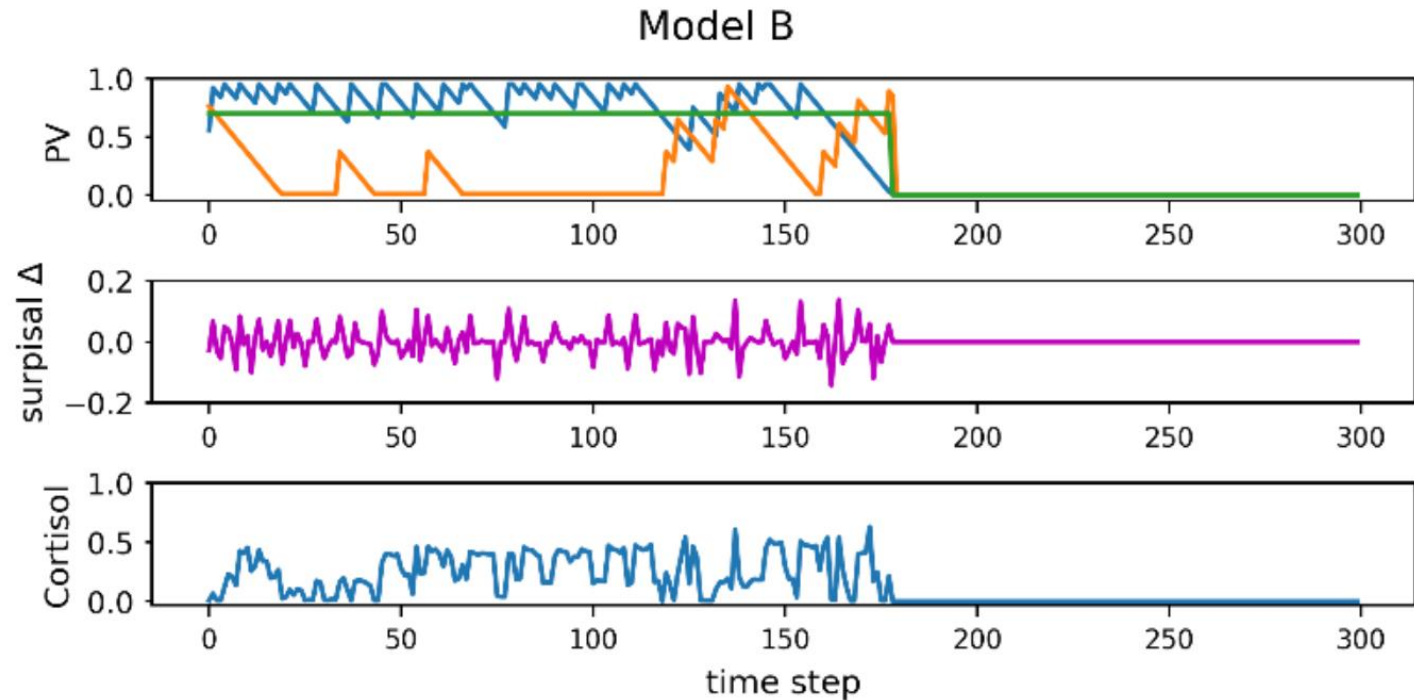
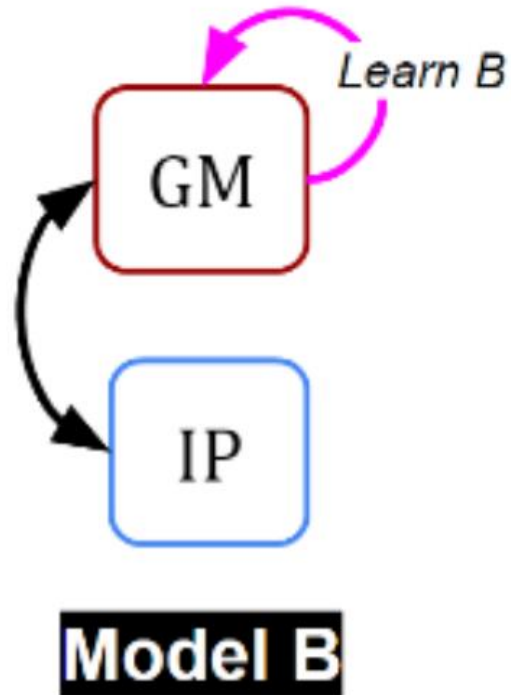
Utilizes active inference to infer motivational states and optimal actions for maintaining homeostasis. solely on homeostatic regulation without learning components.

Exhibits significant fluctuations in internal variables and lower median comfort. Cortisol levels remain relatively stable.



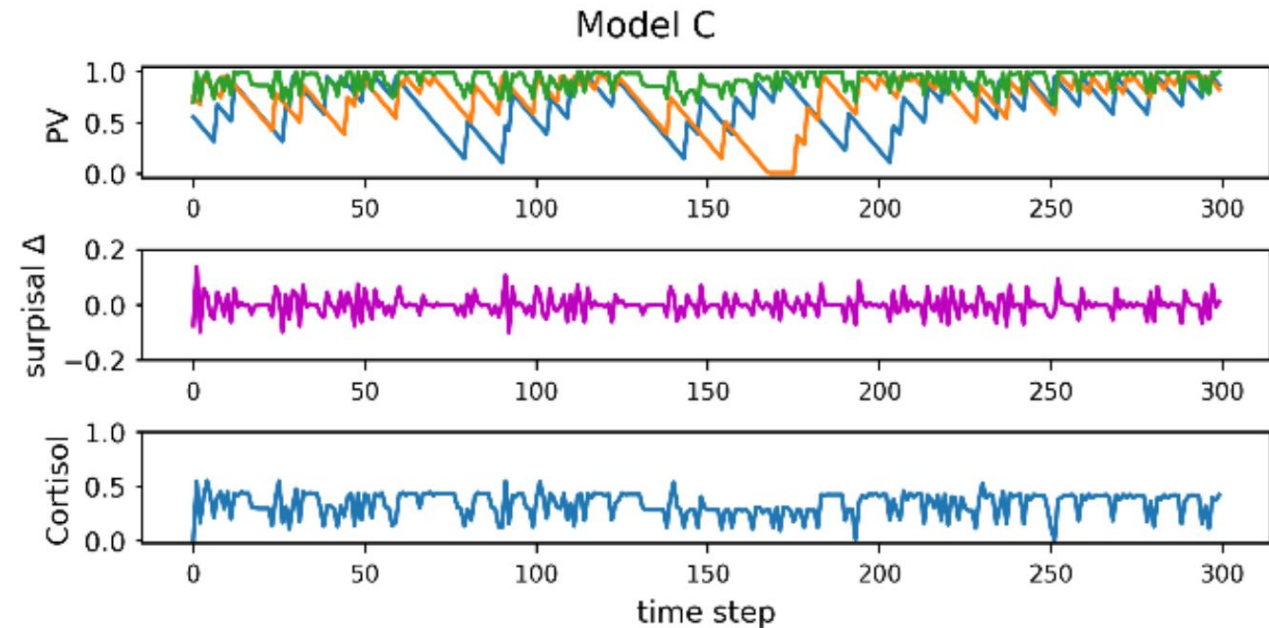
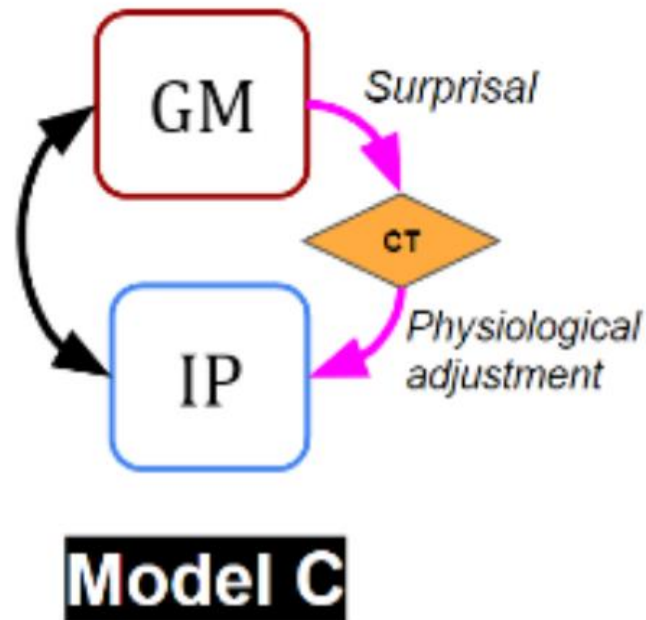
Homeostatic Regulation with Learning

Shows improved regulation of internal variables due to the learning component, resulting in higher median comfort.



Homeostatic Regulation with Cortisol Modulation

Incorporates cortisol modulation, improving energy management and overall viability.



Full Allostatic Regulation with Cortisol Modulation

Further improves performance by integrating cortisol's effect on learning rates, leading to the highest viability and adaptive regulation of internal states.

